**MASTER'S THESIS — COMPLETE THREE-CHAPTER RESEARCH MANUSCRIPT**

CNN-Based Recognition of Handwritten Letters and Evaluation of Writer Attributes

Literature Review • System Design • Experimental Evaluation

Prepared as an editable academic research draft · October 2026

Academic integrity notice: This manuscript is a research-design thesis draft, not a claim that software has been trained or experiments have been performed. All quantitative result fields are explicitly marked NOT YET MEASURED. Replace them with verifiable experimental evidence before submission. The proposed mood/personality prediction tasks are exploratory, not validated measurements of people.


## Abstract

This thesis investigates offline handwritten letters as visual inputs to an artificial intelligence system. It distinguishes transcription of text from statistical estimation of author attributes and tests whether additional predictions about self-reported personality and contemporaneous mood can outperform defensible baselines. The proposed system uses document preprocessing, a convolutional visual encoder, a sequence-recognition branch trained with connectionist temporal classification, and separate heads for gender labels, age bands, and writing-hand labels where suitable consented metadata exist. The study specifies subject-disjoint training and evaluation, preprocessing ablations, confidence calibration, cross-dataset tests, subgroup audits, privacy controls, and a prospective protocol for independently obtaining questionnaire-based psychological ground truth. Because no experiment logs or collected questionnaires were supplied, the results chapter supplies complete analysis plans and publication-ready reporting templates without inventing outcomes. The expected contribution is a reproducible prototype and a scientifically bounded answer to which handwriting-derived traits, if any, generalize beyond confounds.


## Research questions

RQ1: Can an image-to-sequence CNN-based model transcribe offline handwritten letters with useful word and character error rates?

RQ2: Under writer-disjoint conditions, how well can handwriting images predict separately provided age-group, gender-label and dominant-hand labels?

RQ3: Does multi-task training improve each endpoint compared with task-specific architectures?

RQ4: Do handwriting features predict independently measured mood or personality beyond chance and confounding controls?

RQ5: Are results calibrated, robust to image acquisition, equitable across subgroups, and reproducible?

## Chapter map and editorial convention

Chapter 1 occupies 30 thematic manuscript units; Chapter 2 occupies 30; Chapter 3 occupies 20. Every unit begins on a new page to preserve the requested 30/30/20 three-stage allocation. The cover, abstract and bibliography may increase the total physical page count. University formatting rules can change pagination.

## 1.01 Research setting

A handwritten letter simultaneously carries linguistic content and traces of an individual writing process. These two signals motivate distinct modeling targets: character strings and attributes attached to a writer. A page may contain several lines, ornaments, deletions, marginal notes and variable illumination.

Research rationale. The review therefore treats image acquisition, recognition, metadata inference and psychologically sensitive claims as different research streams instead of collapsing them into generic “handwriting analysis.”

Validity and reproducibility. The manuscript must not confuse recognizing what a person wrote with diagnosing who that person is. De-identified datasets can support transcription even when person-level metadata are unavailable.

Required evidence. State the observable target, dataset source, label definition and evaluation unit for every proposed output.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.02 Problem definition

Offline handwriting recognition maps a rasterized letter image into a transcription. Writer-attribute estimation maps an image or group of image patches into a metadata variable measured independently of the image. The problem is inherently multi-output but labels may not be available jointly.

Research rationale. Represent an image by X, its transcription by Y, and independently collected writer labels by A, G, H, P and M. The system approximates conditional distributions P(Y|X) and P(A,G,H,P,M|X) through separate heads.

Validity and reproducibility. Correlation between visual patterns and any label does not establish a biological mechanism or a legitimate individual judgment; partial labels require masks in the loss.

Required evidence. Distinguish the model's learning objective from the interpretation of predictions.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.03 Scientific questions

An authentic contribution cannot be defined simply as deploying a CNN. Recognition has extensive precedent, whereas prediction of stable psychological traits from penmanship is scientifically disputed. The study must test separable falsifiable claims.

Research rationale. Pre-register transcription and demographic-model comparisons. For personality and mood, set a conservative null hypothesis of no generalizable benefit over a demographic, text-content and majority-class baseline.

Validity and reproducibility. Selection bias, writer identity leakage, language and prompt content can produce apparent predictive performance that disappears on independent subjects.

Required evidence. Produce a testable hypothesis table and evidence required to reject the null.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.04 Online versus offline acquisition

Online handwriting records pen trajectory, temporal order, velocity and sometimes stylus pressure. Offline handwriting is a static image. The distinction affects inference about movement and pressure.

Research rationale. For a scanned letter, measure only observable pixel features unless calibrated writing instrumentation provides additional measurements. Distinguish stroke darkness from mechanical pen pressure.

Validity and reproducibility. An offline image cannot recover time-varying pen dynamics uniquely; several movements can produce similar ink patterns.

Required evidence. Describe exactly which quantities are measured and which are unknown.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.05 Recognition history

Classical character-recognition systems used thresholding, projection profiles, contour descriptors, handcrafted features and classifiers. Recognition of cursive writing is difficult because adjacent characters overlap and segmentation boundaries are ambiguous.

Research rationale. Compare legacy pipelines against image-to-sequence recognition, noting why line-level transcription can avoid explicit character segmentation. Plamondon and Srihari reviewed both online and offline approaches [1].

Validity and reproducibility. Historical comparisons should account for different datasets and evaluation units rather than declaring a newer method universally superior.

Required evidence. Record baseline assumptions and citations for each technological generation.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.06 Feature engineering

Local gradients, zoning, skeletons, ascender/descender proportions and connected components provide interpretable descriptors of script geometry. Such features can support classical baselines and diagnostic analysis.

Research rationale. Evaluate a histogram-of-oriented-gradients plus SVM or random-forest baseline on the same subject-disjoint split as the neural model.

Validity and reproducibility. Features can encode scanner resolution, page rulings or language instead of writer attributes.

Required evidence. Report preprocessing and hyperparameters so that the baseline remains competitive.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.07 Image quality

Resolution, motion blur, paper texture and illumination affect thin strokes and punctuation. The same writer can appear stylistically different across pens or capture devices.

Research rationale. Test normalization, illumination correction, noise filtering and geometric rectification as independently controlled preprocessing steps.

Validity and reproducibility. Aggressive denoising erases pen endpoints; binarization may remove low-contrast strokes and erase useful gray-level cues.

Required evidence. Construct an image-quality audit with controlled corruption tests.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.08 Segmentation levels

Whole-page, line, word and patch representations answer different questions. A line gives the transcript model horizontal sequence structure; multiple patches supply variability estimates for writer-level prediction.

Research rationale. Retain page-level provenance and writer identifiers, segment lines for transcription, and sample non-overlapping patches for training with aggregation at the writer level.

Validity and reproducibility. Treating thousands of patches as independent subjects inflates sample size and confidence intervals.

Required evidence. State whether evaluation occurs per character, line, page or writer.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.09 CNN fundamentals

Convolutional neural networks learn localized filters with shared weights, enabling hierarchical visual descriptors. Early layers detect edges; deeper layers combine curves, junctions and spatial patterns.

Research rationale. For feature map h, a convolutional filter computes h_out(i,j,k)=sigma(b_k+sum_{u,v,c} W(u,v,c,k)h_in(i+u,j+v,c)).

Validity and reproducibility. Translation equivariance alone does not guarantee invariance to skew, writer style or context, and pooling trades spatial precision for robustness.

Required evidence. Identify kernel sizes, strides, residual connections and normalization choices.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

| Evaluation component | Required disclosure |
| --- | --- |
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 1.10 Optimization concepts

CNN parameters are typically optimized by stochastic gradient variants using a training set and a separately held-out validation set. Learning-rate schedules, normalization and regularization influence stability.

Research rationale. Specify AdamW or a comparable optimizer, a prespecified learning-rate search grid, gradient clipping and early-stopping rules.

Validity and reproducibility. An architecture comparison without equal tuning budgets can reward the model given more experimentation rather than the better model.

Required evidence. Report fixed seeds, hardware, software versions, search budgets and checkpoints.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.11 Residual encoders

Residual connections facilitate optimization of deeper networks and have been widely used for image representation learning [5]. They are candidates for a shared backbone in handwriting analysis.

Research rationale. Compare a compact custom CNN and a residual CNN initialized either randomly or with available transfer weights. Match input resolution and downstream heads.

Validity and reproducibility. Transfer learning from natural images may not benefit every script, and stronger encoders may overfit limited writer labels.

Required evidence. Use parameter counts, inference time and confidence intervals alongside predictive metrics.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.12 Sequence learning

Transcription cannot generally be modeled as one label for an entire page. A sequence learner transforms image columns or learned time steps into probabilities over characters plus a blank token.

Research rationale. A CNN–BiLSTM–CTC baseline uses convolutional features, bidirectional context and connectionist temporal classification to handle unsegmented text [3].

Validity and reproducibility. CTC conditional independence and repeated-character behavior should be examined rather than hidden behind aggregate CER.

Required evidence. Define the input-height normalization, vocabulary and blank-token rules.

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## 1.13 CTC objective

Connectionist temporal classification sums the probabilities of alignments that collapse to the same target string. It avoids the need for character-level boundary annotations.

Research rationale. For target y and frame predictions p_t, define L_CTC=-log sum_{pi:B(pi)=y} product_t p_t(pi_t|X).

Validity and reproducibility. A model that sees transcriptions of the test lines during language-model selection introduces evaluation leakage.

Required evidence. Use exact definitions of normalization and train/validation/test text separation.

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| **Result status** | NOT YET MEASURED |

## 1.14 Recognition evaluation

Character error rate measures the normalized edit distance between predicted and reference character sequences; word error rate applies the same logic to whitespace-delimited words.

Research rationale. CER=(S_c+D_c+I_c)/N_c and WER=(S_w+D_w+I_w)/N_w. Retain punctuation and case policies consistently.

Validity and reproducibility. Results change with tokenization, transcript normalization and handling of illegible words.

Required evidence. Publish examples of insertion, deletion and substitution errors with counts.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

| Evaluation component | Required disclosure |
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| **Primary unit** | Writer for traits; line or character for OCR |
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| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 1.15 IAM dataset

The IAM corpus is a widely used collection of scanned English handwriting with line and word transcription annotations and writer information [2]. It offers a practical baseline for English offline text recognition.

Research rationale. Follow an official or documented partition, preserve provenance and check license conditions before redistribution. Writer-disjoint splitting should be asserted programmatically.

Validity and reproducibility. IAM cannot establish validity for mood or personality, which are not automatically present as validated questionnaire labels.

Required evidence. Document the exact release, training subset and writer partition.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.16 KHATT dataset

The KHATT corpus provides Arabic handwriting and has been used for demographic handwriting experiments [6], [7]. It offers a cross-script challenge rather than a direct replacement for English letter data.

Research rationale. Compare within-language experiments and separately evaluate cross-language representation transfer. Retain metadata labels as defined by the corpus documentation.

Validity and reproducibility. Arabic and Latin scripts differ in stroke connectivity, writing direction and orthographic structure; cross-dataset shifts confound naïve comparisons.

Required evidence. State language, source, metadata availability and split compatibility.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.17 CVL and other corpora

Additional datasets such as CVL contain writer-indexed handwriting and can probe external validity. The value of a second corpus is not merely its image count but the difference in subjects and acquisition conditions.

Research rationale. Create a dataset-compatibility matrix with script, writer count, labels, image resolution, access conditions and prohibited uses.

Validity and reproducibility. Do not merge demographic categories whose meanings or collection methods differ without justified recoding.

Required evidence. Record the source URL, license and preprocessing exceptions for each dataset.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.18 Gender label interpretation

Published work demonstrates handwriting-based gender-label classification experiments, including CNN-based research by Morera and colleagues [6]. Such labels may reflect a dataset questionnaire or simplified demographic categories.

Research rationale. Treat the variable as a self-reported dataset label where available and give the exact operational categories and missing/other rules.

Validity and reproducibility. Gender identity cannot be determined reliably from handwriting and binary corpus labels do not encompass all identities.

Required evidence. Report balanced accuracy and macro-F1, avoid claims of biological or forensic identification.

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## 1.19 Handedness prediction

Dominant hand can affect motor habits, but writing tools, training, posture and compensation strategies may obscure visual cues. Handedness classification is empirically investigable when labels are independently measured.

Research rationale. Define left-handed/right-handed/ambidextrous labels before training; consider restricting evaluation to reliably annotated cases.

Validity and reproducibility. A right-handed person can learn to write with the left hand, and the writing hand used for a particular letter may differ from preferred hand.

Required evidence. Report baseline prevalence and uncertainty and separate dominant hand from hand used on the sample.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.20 Age estimation

Handwriting changes across development and education, but chronological age is not directly encoded in a letter. Age-group tasks typically have noisy boundaries and can reflect cohort, script or schooling.

Research rationale. Define age bins a priori, e.g., 18–29, 30–44, 45–59 and 60+, if enough samples support those bins; otherwise revise before analysis.

Validity and reproducibility. A heavily imbalanced set can yield misleading apparent accuracy from always predicting the largest group.

Required evidence. Report per-bin recall, confusion matrices and calibration.

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## 1.21 Multi-task learning

Related prediction tasks may share low-level image representations but pull higher layers toward different information. Transcription can encourage stylistic invariance while writer profiling benefits from retaining some style.

Research rationale. Compare fully independent models, one encoder with task heads, and partially shared encoders with separate high-level branches.

Validity and reproducibility. Negative transfer must be quantified task by task and cannot be diagnosed from aggregate multi-task loss alone.

Required evidence. Track task-specific validation curves and an ablation of shared versus independent features.

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## 1.22 Data leakage and split design

Writer-specific visual signatures make random image-level splits especially dangerous. The same author's passages could appear in both training and testing, making memorization look like generalization.

Research rationale. Create subject-disjoint partitions at the earliest preprocessing stage, before crops and augmentations; assert set(train_writer)∩set(test_writer)=empty.

Validity and reproducibility. Repeated collection sessions and duplicate scans can leak through identifiers even when file names differ.

Required evidence. Audit duplicate hashes, collection sessions and partition integrity.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

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|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 1.23 Personality constructs

Personality research usually operationalizes relatively stable traits through validated instruments, such as Big Five questionnaire scales. Handwriting appearance is not itself a validated personality measurement.

Research rationale. Predefine the questionnaire, validated language version, scoring procedure, consent and time interval between questionnaire and letter.

Validity and reproducibility. Graphological impressions are unsuitable as training ground truth because they recycle the disputed claim rather than test it.

Required evidence. Assess incremental prediction of real questionnaire scores and null results.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.24 Graphology validity

Dazzi and Pedrabissi found that graphological assessments did not establish validity for Big Five traits and observed low inter-rater reliability [8]. The limitation persists regardless of whether the predictor is a person or a CNN.

Research rationale. Compare a visual model with an intercept baseline, demographic covariates and the letter's semantic-content model.

Validity and reproducibility. If content or demographic data explains associations, image-based psychological conclusions are not justified.

Required evidence. Report effect sizes, confidence intervals, correction for multiple traits and independent replication.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

| Evaluation component | Required disclosure |
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| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 1.25 Mood constructs

Mood is a context-dependent psychological state that may vary across days or hours. A single letter image does not inherently specify the writer's emotional experience.

Research rationale. Collect a validated self-report mood measure immediately before or after writing, along with timestamp, prompt, pen, device and collection context.

Validity and reproducibility. A classifier may learn emotional words in the letter rather than penmanship; it must be evaluated against a text-only comparison.

Required evidence. Use contemporaneous ground truth and avoid clinical language or diagnostic predictions.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.26 Fairness and bias

Handwriting datasets may over-represent particular ages, education levels, genders, languages, physical abilities or regions. Accuracy estimated in one sample can hide failures in others.

Research rationale. Compute subgroup performance only where groups have sufficient counts and consent. Use macro metrics and bootstrap intervals instead of relying on overall accuracy.

Validity and reproducibility. Publishing small-cell demographic results may expose participants; categories should not be inferred or guessed to fill missing metadata.

Required evidence. Include a disclosure and subgroup-size suppression policy.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.27 Privacy and consent

A handwritten letter can expose identity through signature, addresses, names, personal disclosures and unique writing style. Visual models trained on such materials carry confidentiality risks.

Research rationale. Use de-identification, encrypted storage, consent scopes, access restrictions, a deletion plan and research ethics approval where required.

Validity and reproducibility. Removing names from transcripts does not necessarily anonymize handwriting biometrics or image backgrounds.

Required evidence. Keep a data-governance register and document participant rights.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.28 Model uncertainty

Softmax output can be overconfident on unfamiliar writers. A prediction is meaningful only in relation to the population, label quality, and evaluation distribution.

Research rationale. Measure calibration with reliability diagrams and expected calibration error; consider temperature scaling on validation data.

Validity and reproducibility. Confidence scores are not probabilities of a person's true identity or personality unless assumptions and calibration are satisfied.

Required evidence. Design abstention behavior for low quality or out-of-distribution input.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.29 Related-work synthesis

The most established evidence supports transcription from handwriting images, while CNN-based demographic classification has been investigated with limitations. The literature does not validate graphology as a personality test.

Research rationale. Organize comparison by task, annotation source, split protocol, metric and external validation rather than collecting incomparable headline percentages.

Validity and reproducibility. Cross-paper numerical rankings are misleading when tasks, corpora and label definitions differ.

Required evidence. Construct a study matrix listing [1]–[10], experimental controls, evidence quality and research gaps.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 1.30 Research gap and objectives

The thesis gap lies in transparent, writer-disjoint evaluation of a unified transcription-and-attribute system and in separating exploratory psychology from established recognition tasks. A credible thesis may establish that some proposed targets are not predictably encoded.

Research rationale. Set objectives for baseline reproduction, modular multi-task training, cross-source robustness, uncertainty assessment and a prospective independently validated psychology study.

Validity and reproducibility. Absence of ground truth is not an accuracy score of zero, and no evaluation can be claimed until actual experiments are run.

Required evidence. Finish Chapter 1 with explicit hypotheses H1–H5 and measurement conditions.

In methodological terms, each claim is conditioned on an observable label, a defined population and a locked evaluation protocol. This distinction is essential when moving from transcription to writer profiling: the available pixels cannot establish that an output category is an intrinsic feature of the person. Results should be phrased in terms of measured generalization rather than individual certainty.

## 2.01 Requirements and scope

The prototype accepts a user-supplied scanned letter, returns recognized text, and optionally displays research-only attribute estimates when a properly validated model exists. The software must refuse unsupported outputs rather than quietly invent them.

Research rationale. Divide requirements into functional modules, error handling, consent safeguards and evaluation needs. Assign every output a model version and data lineage.

Validity and reproducibility. A research demonstration must not present demographic predictions as verified facts about an individual.

Required evidence. Produce a requirements traceability matrix.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.02 Use cases and roles

The primary actor is a consenting researcher or participant uploading their own letter. A researcher can label data, train models, view aggregate statistics and export anonymous experiment reports.

Research rationale. Define separate privileges for participant, data curator and administrator and a clear audit trail for image access.

Validity and reproducibility. A demo that accepts anonymous third-party letters without consent invites profiling misuse.

Required evidence. Write acceptance tests for each actor and permission boundary.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.03 System context

The architecture separates frontend, API, preprocessing service, model inference service, metadata store and experiment tracking. This decoupling allows reproducible retraining without coupling results to interface code.

Research rationale. Specify request IDs and immutable model artifact hashes; return processing errors with safe, nonidentifying messages.

Validity and reproducibility. Persisted user images and temporary inference artifacts should have enforced retention periods.

Required evidence. Provide a component diagram and interface contract.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

### 2.04 Upload and validation

Input can be JPEG, PNG or lossless TIFF with finite maximum size and expected minimum resolution. Malformed images should be rejected before inference.

Research rationale. Validate MIME signature and decoded dimensions; remove unneeded metadata, reject decompression bombs and log only non-sensitive operational events.

Validity and reproducibility. Server logs must not accidentally store names, OCR text or raw letter contents.

Required evidence. Test corrupted, rotated, oversized and low-resolution images.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.05 Image preprocessing

Perform orientation estimation, correction of perspective, contrast adjustment and, when useful, background normalization. Preserve a copy of the unmodified input for controlled comparisons.

Research rationale. Each operation must be deterministic in inference; record transform settings to reproduce results.

Validity and reproducibility. Normalization designed for transcription can remove darkness patterns that an attribute model attempts to analyze.

Required evidence. Evaluate per-task preprocessing variants rather than enforcing one irreversible pipeline.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.06 Segmentation

Line segmentation is needed for the CTC branch; attribute inference may instead use tiles sampled across a page. Connected components and projection profiles provide interpretable baselines.

Research rationale. Use layout-aware reading order and reject empty lines; associate every crop with source, writer and page identifiers.

Validity and reproducibility. Text from overlapping columns or marginal notes can be incorrectly concatenated.

Required evidence. Develop line-boundary quality checks and annotated examples.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.07 Data schema

Separate image objects, writer records, sessions, labels, transcripts and consent agreements. Do not store inferred labels as though they were ground truth.

Research rationale. Use stable pseudonymous writer IDs; fields should indicate provenance, measurement instrument and missingness.

Validity and reproducibility. Unrecorded label provenance makes supervised model results scientifically uninterpretable.

Required evidence. Write a versioned data dictionary with schemas and constraints.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

| Evaluation component | Required disclosure |
| --- | --- |
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 2.08 Dataset preparation

Public recognition corpora and independently collected metadata corpora require different inclusion criteria. The prototype should support partial supervision without forcing every sample into every task.

Research rationale. Maintain manifest rows with checksum, writer ID, split, label availability and preprocessing version.

Validity and reproducibility. Mixing datasets can introduce acquisition shortcuts, especially when class labels are source-specific.

Required evidence. Publish counts per split, source, task and unique writer.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.09 Consent-based collection

Personality and mood cannot be evaluated from the intended public recognition corpora without independent measurements. A prospective protocol must collect consent, standardized writing samples and questionnaires.

Research rationale. Recruit a diverse adult sample, counterbalance writing prompts and pen/device conditions, and preregister exclusion criteria.

Validity and reproducibility. Psychological questionnaires or personal letters can contain sensitive information, requiring review and restricted access.

Required evidence. Prepare participant information, withdrawal process and collection forms.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.10 Label policies

Label definitions must correspond to measured constructs: exact or age-band at collection, self-reported gender label, stated dominant writing hand, questionnaire score and state-mood scale.

Research rationale. Retain missing or declined responses without coercion; encode category mappings consistently.

Validity and reproducibility. Age bins and demographic categories can be too sparse for responsible per-class reporting.

Required evidence. Publish a label-availability and eligibility matrix.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.11 Split procedure

The atomic evaluation unit for writer-attribute prediction is the person, not the patch. Split writers first and derive all page and patch assignments from that membership.

Research rationale. Reserve untouched test writers before architecture selection; use a separate validation set and stratification where feasible.

Validity and reproducibility. A train/test overlap by author invalidates claims about predictions for unseen people.

Required evidence. Include executable split assertions and a subject-count report.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 2.12 Augmentation

Training-time rotation, affine distortion, illumination shifts and mild noise can improve robustness; augmentations must not change text or labels.

Research rationale. Apply random transforms only after splitting and keep an audit log of augmentations.

Validity and reproducibility. Mirroring images can alter script direction and obscure left/right semantic assumptions.

Required evidence. Compare clean training with constrained augmentation in an ablation.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.13 Shared encoder

The principal feature extractor is a compact residual CNN that converts an input tensor into spatial activation maps. Separate pathways use these activations differently.

Research rationale. For transcription, collapse vertical dimensions into ordered column features. For demographics, aggregate regions through attention or average pooling.

Validity and reproducibility. A single latent representation can create negative transfer between style-invariant OCR and style-sensitive inference.

Required evidence. Report tensor shapes, layers, parameter counts and compute budget.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.14 Recognition decoder

An OCR head converts sequential features to a distribution over characters plus blank. A BiLSTM captures left and right context; CTC learns alignment to the reference line.

Research rationale. Define training data normalization, alphabet including punctuation, decoding beam width and optional language-model usage.

Validity and reproducibility. A lexical decoder may conceal visual errors and inflate performance if it was tuned on held-out transcripts.

Required evidence. Output raw and decoded predictions for error analysis.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.15 Gender label head

A classification head maps pooled features to dataset-defined categories where consented gender labels are available. Inference must be marked as an unverified statistical estimate.

Research rationale. Use weighted cross-entropy only after analyzing imbalance and compare with macro-metric-aware baselines.

Validity and reproducibility. Dataset labels may be restricted, obsolete or not representative of gender diversity.

Required evidence. Report per-category metrics and appropriate abstention.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.16 Age-group head

Age-group estimation uses categorical cross-entropy with predefined bins or ordinal methods when support allows. Exact age regression is not assumed reliable.

Research rationale. Compare categorical and ordinal objectives and use ordinal distance metrics if grades are ordered.

Validity and reproducibility. Age label may correlate with acquisition year or handwriting instruction instead of direct physical traits.

Required evidence. Report balanced accuracy, per-bin recall and ordinal error.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.17 Handedness head

Handedness is modeled only when the hand used to produce the sample or the writer's reported preference is known, distinguishing the two concepts.

Research rationale. Use an independent head with subject-level loss; patches from one letter contribute an aggregated writer-level prediction.

Validity and reproducibility. Ambidexterity, injury and adapted writing styles complicate simplistic binary classification.

Required evidence. Include confusion matrices and a statement of the label's operational meaning.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.18 Personality exploratory head

A research-only head may regress validated Big Five scale scores, but it cannot be enabled for user-facing judgments without evidence of transportable validity.

Research rationale. Use a single preregistered questionnaire and regression metrics, comparing handwriting imagery to demographic and text-based predictors.

Validity and reproducibility. Text content or population selection can create apparent links unrelated to handwriting form.

Required evidence. Evaluate incremental $R^2$ with confidence intervals and multiplicity correction.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.19 Mood exploratory head

Mood estimation is conditional on a contemporary self-report. A model cannot infer clinical state from a written page without validated evidence.

Research rationale. Predefine questionnaire timing, assessment scale, prompt and collection controls; use regression or predefined threshold classification.

Validity and reproducibility. Transient mood can change during a long writing session and self-report measurement itself has noise.

Required evidence. Compare image-only, text-only and multimodal baselines.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.20 Multi-task loss

A combined loss can optimize OCR and available classification/regression targets with masks for missing supervision. It must prevent large-scale CTC gradients from dominating all heads.

Research rationale. Let $L=\lambda_o L_{CTC}+\Sigma_k m_k \lambda_k L_k$ where $m_k \in \{0,1\}$ indicates a recorded label. Tune $\lambda$ only using training/validation data.

Validity and reproducibility. Naïve loss weighting may favor tasks with more samples or stronger signals; full supervision is not guaranteed.

Required evidence. Record per-task loss weights and learning curves.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 2.21 Training pipeline

Reproducible runs require pinned package versions, deterministic data manifests, experiment IDs, seeds, checkpoint selection and validation-triggered stopping.

Research rationale. Use a configuration file with optimizer, batch sizes, schedules, encoder choice, preprocessing hashes and checkpoint artifact.

Validity and reproducibility. Validation-set overuse yields optimistic estimates when dozens of design decisions are tuned informally.

Required evidence. Publish a run ledger with hyperparameter and dataset revisions.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.22 Hyperparameter search

Hyperparameters include width and height normalization, encoder depth, dropout, optimizer, learning rate, class weights and loss coefficients.

Research rationale. Fix a finite search budget and select by a predefined primary metric. Never use the final test split for iteration.

Validity and reproducibility. One-number comparisons are unstable on small writer populations.

Required evidence. Report search space, selection rule, repetitions and variability.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.23 Inference service

The serving application accepts an image, segments and recognizes text, aggregates patch logits and returns supported outcomes with model version and calibrated uncertainty.

Research rationale. Implement explicit feature flags for unvalidated heads and a “not supported” response if a compatible model is not deployed.

Validity and reproducibility. Output formatting must not portray low-confidence or research-only estimates as factual conclusions.

Required evidence. Specify example JSON schema and response validation.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 2.24 Frontend design

The interface displays the uploaded page, corrected OCR, image quality warnings, and explanation of model uncertainty. It uses plain language about experimental targets.

Research rationale. Provide clear separate panels for recognized text, evaluated metadata experiments and unsupported experimental proposals.

Validity and reproducibility. Presenting personality or mood in a dashboard can cause unjustified confidence even if caveats appear elsewhere.

Required evidence. Perform usability tests for calibration and interpretability.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.25 Security controls

Letters may contain personal information. Enforce transport encryption, access control, least-privilege object storage, file scanning and retention enforcement.

Research rationale. Store consent records separately from images, avoid direct identifiers in training manifests and encrypt model-related archives as needed.

Validity and reproducibility. Training artifacts may memorize rare textual strings; secure release requires review and sample-level privacy analysis.

Required evidence. Provide a threat model and access-control tests.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 2.26 Quality assurance

Unit tests should cover decoding, preprocessing, data types, split assertions and label masking. Integration tests should exercise uploads and inference service failure paths.

Research rationale. Test reproducibility across two fresh environments, verify batch and single-sample consistency, and compare deterministic checkpoints.

Validity and reproducibility. Silently skipped corrupted images may bias evaluation if failures concentrate in a subgroup.

Required evidence. Publish pass/fail counts and fault handling criteria.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.27 Observability

Operational metrics include request latency, image rejection rate, preprocessing failures and inference memory. Research metrics include drift indicators and calibration on a consented evaluation sample.

Research rationale. Log aggregate service events without storing private text or images and use privacy-preserving quality summaries.

Validity and reproducibility. Production drift monitoring alone does not establish that demographic or psychological predictions remain ethically justified.

Required evidence. Define monitoring thresholds and incident response.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.28 Deployment and reproducibility

Containerization simplifies environment parity; artifact storage provides traceable model weights and manifests. The design should be deployable locally for a university demonstration.

Research rationale. Document environment variables, dependency locking, startup steps and a reproducibility checklist.

Validity and reproducibility. A working UI is not proof of a scientifically valid model.

Required evidence. Provide installation, test and evaluation instructions.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.29 Integration acceptance criteria

A system is complete only when every component can be exercised against its documented contract, including preprocessing, segmentation, versioned model inference, consent guards and error handling.

Research rationale. Run an end-to-end evaluation with a known handwritten page and assert response structure, latency logging and permission boundaries.

Validity and reproducibility. A demonstration on one attractive sample cannot establish generalization, model accuracy or privacy compliance.

Required evidence. Produce a signed acceptance checklist and reproducible walkthrough.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 2.30 Chapter synthesis

The proposed architecture intentionally decouples OCR from inference claims and guards psychology tasks behind validated independent evidence. The design permits negative as well as positive findings.

Research rationale. The next chapter defines subject-level experiments, statistical comparisons, model cards and reporting conventions before observing outcomes.

Validity and reproducibility. Architectural elegance cannot replace ground-truth quality, external replication, fairness review or calibrated uncertainty.

Required evidence. Conclude with an implementation readiness checklist and outstanding dependencies.

Implementation note. Module contracts must remain independent of dataset-specific labels, and every configuration change should be versioned. The retained experiment manifest should be sufficient for another researcher to reconstruct train/validation/test membership, augmentation rules, model weights and the final scoring script.

## 3.01 Study design

The empirical program is staged so that transcription is evaluated first, attribute classification second, and personality/mood only if appropriate participant data have been collected.

Research rationale. Use preregistered hypotheses, writer-disjoint splits, baseline comparisons and a single held-out test evaluation per frozen model.

Validity and reproducibility. No trained model logs or participant-level data were supplied; therefore no numeric performance claim can be made in this manuscript.

Required evidence. Report sample counts, exclusion flows and timestamps from actual execution.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

## 3.02 Experimental environment

Hardware and software versions materially affect reproducibility, even when final predictions should be stable.

Research rationale. Record GPU or CPU model, RAM, operating system, Python, PyTorch, OpenCV, CUDA if used and random seeds.

Validity and reproducibility. Planned hardware is not a measurement; do not write “trained on GPU X” until training occurs.

Required evidence. Environment table: hardware=NOT YET RECORDED; versions=NOT YET RECORDED.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

### 3.03 Dataset census

A meaningful result begins with people and pages represented in each split, not a total image count inflated by crops.

Research rationale. Count unique writers, letters, lines, patches, age bins, hand preference groups and missing labels per dataset and partition.

Validity and reproducibility. Patch count must not be used as the denominator for writer-level confidence intervals.

Required evidence. Counts table: writers=NOT YET MEASURED; lines=NOT YET MEASURED; task-eligible cases=NOT YET MEASURED.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

### 3.04 Transcription baselines

Recognition comparisons must include a simple benchmark and a CNN-based sequence model trained under matching split policies.

Research rationale. Report CER, WER, median inference latency and error types for identical test lines.

Validity and reproducibility. A language model might enhance fluent output without improving letter-image recognition; report whether decoding used one.

Required evidence. Table columns: model, CER, WER, mean latency, confidence interval; all entries pending.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

### 3.05 Training curves

Loss trajectories distinguish underfitting, overfitting and optimization failure.

Research rationale. Plot training CTC loss and validation CER against epoch, with early-stopping epoch and chosen checkpoint highlighted.

Validity and reproducibility. The lowest training loss may correspond to worse performance on unseen writers.

Required evidence. Insert actual training curves and saved checkpoint hashes after execution.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

### 3.06 OCR qualitative errors

Confusions in connected letters, diacritics, punctuation and unusual slant are useful for interpreting failures.

Research rationale. Select samples via fixed random seeds plus predefined error categories, not only attractive successes.

Validity and reproducibility. A curated gallery of only strong examples is not representative of failure rates.

Required evidence. Present paired reference/prediction strings with anonymized crops and categorized edit operations.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

### 3.07 Preprocessing ablation

Deskew, binarization, gray-scale input and line-height normalization may help some writers and harm others.

Research rationale. Change one factor at a time using the frozen split, identical training budget and several random seeds.

Validity and reproducibility. Choose variants on validation data; leave the test set untouched until selection is locked.

Required evidence. Report mean CER difference, confidence intervals and visually inspected failure categories.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

### 3.08 Attribute baselines

Gender labels, age bands and handedness must each have a majority-class or stratified baseline and preferably a classical feature-based baseline.

Research rationale. Score predictions per writer after aggregating pages or patches using a prespecified rule.

Validity and reproducibility. Overall accuracy can disguise failure on minority classes and should not be the only metric.

Required evidence. Report macro-F1, balanced accuracy, recall by class and confidence intervals.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

### 3.09 Gender-label evaluation

This evaluation tests a correlation between handwriting features and consented labels in a defined dataset, not the identity of a person.

Research rationale. Provide confusion matrix, balanced accuracy and comparison with prevalence-aware baselines.

Validity and reproducibility. Results must be presented as dataset-specific and should not classify unknown individuals with unwarranted certainty.

Required evidence. All performance cells remain NOT YET MEASURED.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

## 3.10 Age evaluation

Age-group classification must reflect its ordinal target and preselected bin boundaries.

Research rationale. Compare macro-F1, mean absolute bin distance and per-bin recall.

Validity and reproducibility. Collection-year artifacts and educational cohort effects are alternative explanations.

Required evidence. Add subgroup and external-dataset tests before interpreting relationships.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

## 3.11 Handedness evaluation

The outcome depends on whether the metadata represent preference or the actual writing hand.

Research rationale. Stratify or separately evaluate those two labels if both are available.

Validity and reproducibility. Dominance and writing-hand choice are not identical and should not be merged silently.

Required evidence. Report sensitivity and specificity for each annotated category when counts permit.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

## 3.12 Multi-task ablation

The shared model should be compared against independent models under comparable parameter or compute budgets.

Research rationale. Measure task-wise performance changes and uncertainty across repeated seeds.

Validity and reproducibility. A gain in one head does not establish overall superiority if OCR degrades substantially.

Required evidence. Create a Pareto comparison across CER, macro-F1, compute and calibration.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

### 3.13 Uncertainty calibration

Classifier probabilities must be checked empirically before they are shown to users.

Research rationale. Plot reliability diagrams and compute expected calibration error on writer-level outputs; fit temperature on validation only.

Validity and reproducibility. An uncalibrated 0.9 prediction is not a 90% certainty about an individual.

Required evidence. Include accuracy-versus-coverage plots under abstention.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 3.14 Robustness tests

Corruption stress tests assess lighting, blur, page rotation, lower resolution and alternate scanners.

Research rationale. Specify perturbation strengths prior to testing and use the same subject-disjoint test subset across severity levels.

Validity and reproducibility. Synthetic distortions do not substitute for real-world external collection.

Required evidence. Report performance degradation relative to clean-image baseline.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

### 3.15 Cross-dataset transfer

External evaluation estimates whether representations carry beyond one corpus or writing environment.

Research rationale. Train on one source where feasible, harmonize only defensible labels, and test on a second source without tuning to it.

Validity and reproducibility. Script, language and metadata definitions can prevent direct comparison; document non-comparability rather than forcing an aggregate score.

Required evidence. Report absolute metric shifts and population/domain differences.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

## 3.16 Fairness analysis

Performance must be assessed across well-supported participant groups while guarding privacy.

Research rationale. Bootstrap writers, compute subgroup error intervals and apply minimum cell-count reporting rules.

Validity and reproducibility. Small categories can yield highly uncertain estimates and reidentification risk.

Required evidence. Publish a limitations table where a subgroup cannot be reliably evaluated.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

### 3.17 Personality experiment

A scientifically valid experiment uses an independent questionnaire as the outcome, not graphologist labels.

Research rationale. Model Big Five scores using image-only features, text-only features, demographics-only covariates and combined predictors; compare out-of-sample $R^2$.

Validity and reproducibility. Because existing graphology studies have not established predictive validity [8], the default hypothesis is that visual features add little or no reliable prediction.

Required evidence. Report confidence intervals, multiple-comparison adjustments and external replication; all values pending.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

| Evaluation component | Required disclosure |
|---|---|
| **Primary unit** | Writer for traits; line or character for OCR |
| **Data provenance** | Dataset release and consent/label instrument |
| **Leakage test** | No writer overlap among partitions |
| **Primary statistic** | Predeclared task-specific metric |
| **Result status** | NOT YET MEASURED |

## 3.18 Mood experiment

A mood experiment requires a state measurement taken near the writing event and attention to the semantic text.

Research rationale. Predict contemporaneous validated scale scores from image-only and content-only representations using participant-disjoint splits.

Validity and reproducibility. An emotional sentence can reflect a writing prompt rather than the writer's true current mood.

Required evidence. Report sample timing, instrument reliability and incremental predictive value; results pending.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

### 3.19 Statistical inference

Accuracy comparisons need uncertainty intervals at the correct clustering level.

Research rationale. Bootstrap writers, report effect sizes and 95% confidence intervals, and preregister the primary test comparison.

Validity and reproducibility. Overlapping patches violate independent-sample assumptions and can make significance tests anti-conservative.

Required evidence. Include multiplicity controls and an analysis of test-set power.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

## 3.20 Final interpretation

A rigorous thesis distinguishes a working OCR prototype from evidence that personal characteristics are inferable. Null or negative results are legitimate scientific outcomes.

Research rationale. Once real experiments exist, relate results to hypotheses, dataset boundaries, observed biases and limitations, and specify reproducible release artifacts.

Validity and reproducibility. Do not use simulated or illustrative values as empirical findings. Without experimental data, definitive claims remain unresolved.

Required evidence. Submission checklist: trained model, verified splits, results figures, participant ethics, citations and supervisor approval.

Status and interpretation. This subsection intentionally contains no fabricated numerical findings. The final dissertation must replace its pending evidence with actual saved outputs, sample sizes, uncertainty intervals, error analyses and the supervisor-approved interpretation of results.

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Literature source verification links: https://doi.org/10.1109/34.824821 ; https://doi.org/10.1155/2018/3891624 ; https://doi.org/10.2466/PR0.105.F.1255-1268

IMPORTANT: This research manuscript contains a completed chapter structure, explanations, methods and experiment plans, but NOT performed experiments or an evidence-based results discussion. It is not submission-ready until the system is built, experiments are run, outcomes inserted and supervisor formatting approved.

# Technical Supplement: Mathematical Models, Core Algorithms and Reproducibility

This technical supplement supplies numbered equations and pseudocode cross-referenced to the three chapters. It is a proposed system specification, not evidence of executed experiments.

## A.1 Notation and task definitions

Let X denote the handwritten image, y the ground-truth transcript; a, g, h the age-band, self-reported gender and handedness labels; s and q the independently measured personality and mood labels; theta the model parameters; m_t the availability mask. Writer identity must be tracked separately from document identity.

## A.2 Mathematical formulas (organized by thesis topic)

### Equation 01: Image normalization

(1) I_norm(x,y) = [I(x,y) - mu_I] / (sigma_I + epsilon)

Normalize training images using training-set statistics only; epsilon prevents division by zero.

### Equation 02: Binarization

(2) B(x,y) = 1 if I(x,y) < tau; otherwise 0

Threshold tau can be estimated using Otsu's method, and its choice must be fixed by the validation protocol.

### Equation 03: Affine deskew transformation

(3) [x_prime, y_prime, 1]^T = A(theta) [x, y, 1]^T

Estimate a geometric transform from image contours; avoid modifying stroke identity.

### Equation 04: Convolutional feature extraction

(4) Y(i,j,k) = phi( b_k + sum_{u,v,c} W(u,v,c,k) X(i+u,j+v,c) )

The convolution kernel captures local stroke patterns and nonlinear activation phi allows hierarchical representations.

### Equation 05: Convolution spatial dimensions

(5) H_out = floor((H + 2P - D(K - 1) - 1)/S + 1)

Here K is kernel size, P padding, D dilation, and S stride; compute output width analogously.

### Equation 06: Batch normalization

(6) BN(x) = gamma (x - mu_B)/sqrt(sigma_B^2 + epsilon) + beta

Batch statistics are estimated in training; running statistics are used in evaluation.

### Equation 07: Residual learning

(7) h_(l+1) = ReLU( F(h_l; W_l) + h_l )

A skip connection can improve gradient flow when a deeper CNN backbone is used.

### Equation 08: Feature pooling

(8) $z_c = (1/HW) \sum_{i=1}^{H} \sum_{j=1}^{W} F(i,j,c)$

Global average pooling summarizes document or patch features for attribute heads.

### Equation 09: BiLSTM recurrence

(9) $h_t = \text{concat}(\text{LSTM_forward}(f_t), \text{LSTM_backward}(f_t))$

Column-wise CNN features become a sequence; the recurrent model uses context in both directions.

### Equation 10: CTC sequence likelihood

(10) $P(y \mid X) = \sum_{\pi \in B^{-1}(y)} \prod_{t=1}^{T} p_t(\pi_t \mid X)$

B collapses repeated symbols and blank tokens; CTC marginalizes over possible alignments.

### Equation 11: CTC recognition loss

(11) $L_{CTC} = -\log P(y \mid X)$

The transcription loss can be optimized without character-level segmentation labels.

### Equation 12: Categorical cross-entropy

(12) $L_{CE} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} y_{ik} \log(p_{ik} + \varepsilon)$

Used for consenting participants' self-reported gender category, age group and handedness; handle imbalance separately.

### Equation 13: Ordinal age loss

(13) $L_{age} = \sum_{r=1}^{K-1} \text{BCE}( P(a > r \mid X), \mathbb{1}[a > r] )$

Optional ordinal modeling respects ordered age bands without implying exact birth-age inference.

### Equation 14: Class weighting

(14) $w_k = N / (K N_k)$; $L_{weighted} = -\frac{1}{N} \sum_i w_{y_i} \log p_{i,y_i}$

Estimate weights from the training partition only; report unweighted metrics too.

### Equation 15: Masked multi-task loss

(15) $L_{total} = [\sum_t \lambda_t m_t L_t] / [\sum_t \lambda_t m_t + \varepsilon]$

$m_t$ marks whether task t has a valid independently measured label; do not infer missing psychological labels.

### Equation 16: Regularized objective

(16) $J(\theta) = L_{total}(\theta) + \lambda_{reg} ||\theta||_2^2$

Tune regularization using the validation split; keep the test split untouched.

### Equation 17: Adam parameter update

(17) $m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$; $v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$; $\theta_t = \theta_{t-1} - \eta \hat{m}_t/(\sqrt{\hat{v}_t}+\varepsilon)$

Bias-corrected moments $\hat{m}$ and $\hat{v}$ are used by Adam; save optimizer state for reproducibility.

### Equation 18: Learning-rate schedule

(18) $\eta_e = \eta_{min} + 0.5(\eta_{max} - \eta_{min})[1 + \cos(\pi e/E)]$

Cosine annealing is an optional scheduler; log all hyperparameter choices.

### Equation 19: Character error rate

$$CER = (S_{char} + D_{char} + I_{char}) / N_{char} \quad (19)$$

Compute edit distance after a declared Unicode normalization policy.

### Equation 20: Word error rate

$$WER = (S_{word} + D_{word} + I_{word}) / N_{word} \quad (20)$$

Report the tokenization rule, especially for Chinese where word boundaries require segmentation.

### Equation 21: Precision and recall

$$Precision = TP/(TP+FP); \quad Recall = TP/(TP+FN) \quad (21)$$

Use the one-versus-rest formulation for each class; disclose zero-denominator policy.

### Equation 22: F1 and macro-F1

$$F1_k = 2\,P_k\,R_k/(P_k+R_k); \quad MacroF1 = (1/K)\sum_k F1_k \quad (22)$$

Macro averaging reduces the dominance of large classes.

### Equation 23: Balanced accuracy

$$BalancedAccuracy = (1/K)\sum_k TP_k/(TP_k + FN_k) \quad (23)$$

This measure is recommended for imbalanced writer attributes.

### Equation 24: Expected calibration error

$$ECE = \sum_{b=1}^{B} (|M_b|/N)\,|acc(M_b) - conf(M_b)| \quad (24)$$

Predefine bins and evaluate probabilities on writer-disjoint holdout examples.

### Equation 25: Baseline improvement

$$Delta_{metric} = Metric_{model} - Metric_{baseline} \quad (25)$$

Compare with a majority-class and a texture-feature baseline under the identical split.

### Equation 26: Bootstrap confidence interval

$$CI_{95} = [quantile_{0.025}(T_{boot}), quantile_{0.975}(T_{boot})] \quad (26)$$

Resample writers, not individual patches, so interval estimates reflect writer-level dependence.

### Equation 27: Personality regression objective

$$L_{personality} = (1/N_{valid}) \sum_{i:\, m_i=1} ||s_i - \hat{s}_i||_2^2 \quad (27)$$

Scores s must come from independently administered validated questionnaires; failure to exceed baseline is valid evidence.

### Equation 28: Mood classification objective

$$L_{mood} = -(1/N_{valid}) \sum_{i:m_i=1} \sum_k y_{ik} \log(p_{ik} + \varepsilon) \quad (28)$$

Mood labels require contemporaneous consent-based reports; image characteristics alone are not ground truth.

## A.3 Executable algorithm specifications (pseudocode)

### Algorithm 01: Writer-disjoint partitioning

01 Input: images and writer IDs; fix random seed.

02 Group all samples by writer ID.

03 Stratify groups by eligible labels where feasible.

04 Assign each writer to exactly one of train/validation/test.

05 Assert set(train_writer) ∩ set(test_writer) = empty.

06 Freeze IDs and report counts.

### Algorithm 02: Offline image preprocessing

01 Load image and retain an unmodified source copy.

02 Validate image quality, orientation and consent status.

03 Convert to grayscale; estimate background and deskew.

04 Denoise gently; retain thin pen strokes.

05 Resize while preserving aspect ratio, then pad to network input.

06 Normalize using training statistics and record transform settings.

### Algorithm 03: CNN–BiLSTM–CTC inference

01 Input preprocessed line image X.

02 Compute feature tensor F = CNN(X).

03 Collapse vertical axis into time-indexed features f_t.

04 Compute contextual states H = BiLSTM(f_1,...,f_T).

05 Apply softmax over character alphabet plus CTC blank.

06 Decode using greedy or beam search and return text/confidence.

### Algorithm 04: Multi-task supervised training

01 Initialize backbone, task heads and optimizer.

02 For each writer-disjoint training batch, compute shared features.

03 Compute CTC loss only for samples with transcriptions.

04 Compute each attribute loss only where independent labels exist.

05 Weight and normalize active losses; backpropagate and clip gradients.

06 Validate at writer level, checkpoint best prespecified objective.

### Algorithm 05: Age, gender and handedness heads

01 Pool features at a prespecified line/page aggregation level.

02 Pass pooled features to three independently parameterized heads.

03 Use eligible self-reported labels; mask missing or withdrawn labels.

04 Calibrate scores on validation writers, not test writers.

05 Report per-class measures, prevalence and uncertainty.

06 Do not present predictions as verified personal identities.

### Algorithm 06: Exploratory psychological validity test

01 Obtain ethical approval and separate informed consent.

02 Collect validated personality questionnaires and contemporaneous mood scales.

03 Freeze hypotheses, covariates and negative controls in a protocol.

04 Train text-only, image-only, and combined models on identical splits.

05 Evaluate writer-disjoint holdout and bootstrap writer-level intervals.

06 Report null outcomes, subgroup gaps and confounding transparently.

### Algorithm 07: Reproducible evaluation

01 Load the immutable test IDs and selected checkpoint once.

02 Compute CER/WER for transcription, macro-F1 and balanced accuracy for classification.

03 Compute calibration curves and ECE from stored probabilities.

04 For psychology tasks compare to preregistered baseline and null model.

05 Report writer-level confidence intervals and per-group counts.

06 Export configurations, seeds, code revision and failure cases.

### Algorithm 08: Model ablation and error analysis

01 Define ablations: no deskew, no attention, single-task, alternative backbone.

02 Keep data splits and evaluation metrics fixed.

03 Train each condition with multiple recorded seeds.

04 Compute deltas and writer-level confidence intervals.

05 Inspect representative errors, acquisition shifts and missing labels.

06 Distinguish descriptive trends from statistically supported conclusions.

## A.4 Proposed CNN module configuration

The following is a proposed reference configuration to implement and tune; it is not a claim that the network has been trained.

- Input: grayscale line images, height 64 pixels, width aspect-ratio preserved.
- CNN block 1: 3×3 conv 32 channels, BN, ReLU, 2×2 pooling.
- CNN block 2: 3×3 conv 64 channels, BN, ReLU, 2×2 pooling.
- CNN block 3: 3×3 conv 128 channels, BN, ReLU; constrain horizontal downsampling.
- Text recognition: vertical collapse, bidirectional LSTM (256 states per direction), CTC classifier.
- Attribute estimation: global average pooling, 128-unit fully connected layer, independent classification heads.
- Personality and mood: activate only with independent validated scales and ethical approval.

## A.5 End-to-end system flow

1. Input

2. Preprocessing

3. Shared CNN backbone

4. Recognition path: column features → BiLSTM → CTC

5. Attribute path: global pooling → age / gender / handedness

6. Exploratory heads: questionnaire-backed personality / mood

7. Writer-disjoint tests; calibrated reporting

### A.6 Required empirical evidence for final submission

Before submission, replace pending empirical content with actually measured sample and writer counts, class distributions, hardware/software versions, learning curves, ablations, metric tables, 95% confidence intervals, error cases, and baseline comparisons. Never convert placeholders or hypothetical outcomes into measured results.

Foundational sources: LeCun et al. [9] (CNN); Graves et al. [3] (CTC); Shi et al. [4] (sequence recognition); He et al. [5] (residual networks); Morera et al. [6] (attribute prediction); Dazzi and Pedrabissi [8] (personality validity).